

Complex Stress Analysis and Visualization

Assignment #Q: 1

A rectangular steel plate with a central hole is subjected to **plane stress** at a critical point near the edge of the hole, where the measured stress components are:

- $\sigma_x = 85$ MPa (tensile), $\sigma_y = -40$ MPa (compressive), $\tau_{xy} = 55$ MPa.
- $\sigma_x = -85$ MPa (tensile), $\sigma_y = 40$ MPa (compressive), $\tau_{xy} = 55$ MPa.
- $\sigma_x = 85$ MPa (tensile), $\sigma_y = 40$ MPa (compressive), $\tau_{xy} = -55$ MPa.
- $\sigma_x = -85$ MPa (tensile), $\sigma_y = -40$ MPa (compressive), $\tau_{xy} = -55$ MPa.

You are required to perform a **comprehensive stress analysis** using theory, computation, and visualization tools.

Tasks:

1. Theoretical Derivation (Manual):

Derive the general equations for normal and shear stresses on a plane inclined at an angle θ to the x-axis. Clearly state your assumptions and show all steps leading to the transformation equations.

2. Computation and Visualization (Excel or ...):

- Using the given stresses, compute the **normal and shear stresses** acting on planes at **every 5° increment from 0° to 180°**.
- Use **Excel (or another suitable tool)** to plot:
 - Normal stress (σ_θ) vs. angle (θ)
 - Shear stress (τ_θ) vs. angle (θ)
- Combine both into one plot to illustrate how stresses vary with orientation.
- Clearly label the angles where **principal stresses** and **maximum shear stresses** occur.

3. Mohr's Circle Construction:

- Construct **Mohr's Circle** using either manual graphing (scanned and inserted) or computational plotting tools.
- Identify the **principal stresses**, **maximum shear stress**, and the corresponding **plane orientations**.
- Compare your graphical results with those obtained from the stress transformation equations.

Complex Stress Analysis and Visualization

Assignment #Q: 1

4. Interpretation and Discussion (Analytical Report):

- Discuss the **physical meaning** of the angle-stress relationships obtained from your plots.
 - Explain how the stress distribution helps in understanding **material failure** or **yielding** around the hole.
 - Discuss **why the Mohr's Circle** is a powerful visualization tool compared to numerical tables.
 - Reflect on any discrepancies between analytical, tabulated, and plotted results, and explain potential causes.
-

BIT MENG:3111